

ISEN/DAEN 427 Lecture 7

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Intro to Exploratory Analysis of Bayesian Models

BCP Ch. 2



What is Exploratory Analysis in Bayesian Modeling?

- In Bayesian modeling, we don't stop once we've computed a posterior.
- We then perform additional tasks:
 - Check our assumptions
 - Understand predictions
 - Diagnose whether the computation/inference “worked”
 - Compare models
- In other words: there is life *before* and *after* inference.

Understanding Your Assumptions

- Before diving into data, think about the assumptions we embed in the model — especially the priors.
- A very useful tool: the *prior predictive distribution* — sample from the model using only the prior (no data) to see what kinds of outcomes your model thinks are likely.
- By varying the prior (e.g., different σ_σ), we see very different implied outcome-distributions.

Understanding Your Predictions

- After fitting the model (posterior), we want to see how well it predicts the data: *posterior predictive checks*.
- Generate synthetic data from the posterior predictive distribution and compare to the observed data.
- Example: for a Binomial model, compare the number (or proportion) of successes in the real data with those from simulations.
- We can also compute “Bayesian p -values”:
e.g.

$$p_B = P(T_{\text{sim}} \leq T_{\text{obs}} \mid \tilde{Y})$$

where T is some test statistic.

Diagnosing Numerical Inference

- When using methods like MCMC to compute the posterior, we must check if the algorithm “worked” well.
- Key diagnostics:
 - Effective Sample Size (ESS)
 - Potential Scale Reduction Factor $\hat{R}$
 - Monte Carlo Standard Error (MCSE)
 - Trace plots, autocorrelation plots, rank plots
 - Divergences in HMC / sampler warnings
- For example, $\hat{R} \approx 1$ suggests chains have mixed well; low ESS means fewer “independent” draws than nominal.

Model Comparison

- We often build multiple models and need a way to compare them.
- A common measure is the *Expected Log Pointwise Predictive Density*:

$$\text{ELPD} = \sum_{i=1}^n \int p_t(\tilde{y}_i) \log p(\tilde{y}_i | y_i) d\tilde{y}_i$$

- In practice, one can approximate this via methods like Leave-One-Out Cross-Validation (LOO) using importance-sampling: PSIS-LOO.
- Software (e.g., ArviZ) can compare models and rank them by predictive accuracy.

Summary

- Bayesian modeling goes beyond “fit model → get posterior”.
- We must:
 - Examine assumptions (via prior predictive checks)
 - Assess predictions (via posterior predictive checks)
 - Diagnose computation/inference (ESS, $\hat{R}$, etc.)
 - Compare models (ELPD / LOO)
- These steps help ensure that our modeling is robust, reliable, and understandable.

Examples

- Try picking a simple Bayesian model (e.g., coin-flip) and:
 - Choose a prior, simulate from the prior predictive distribution.
 - Fit the model to some observed data.
 - Perform posterior predictive checks: how well does the model capture the data?
 - Use a sampler (software) and look at diagnostics (trace plots, $\hat{R}$, ESS).
 - Build a second model (e.g., with a different prior or additional parameter) and compare via predictive accuracy.
- When might a model *look* good (fit the data) but still be problematic?